

The Nonnegative Heisenberg Kernel Threshold Is Exactly Four

Candidate full counterexample to arXiv:1911.03223

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Status: candidate full counterexample, likely valid. For every $0 < \alpha < 4$ there is a compact regular curve in the first Heisenberg group on which the singular integral associated with K_α is not uniformly L^2 bounded. Hence the positive theorem for $\alpha \geq 4$ is sharp, and in particular it does not persist for $\alpha \geq 2$. Independent expert review is still required.

1 Source question and current boundary

Fässler and Orponen [1] consider the nonnegative homogeneous kernels

$$K_\alpha(x, y, t) = \frac{|t|^{\alpha/2}}{\|(x, y, t)\|^{\alpha+1}}, \quad \alpha \geq 1,$$

on the first Heisenberg group. Their Theorem 1.7 proves boundedness on every regular curve for $\alpha \geq 4$, and the paragraph immediately afterward asks whether the result persists for $\alpha \geq 2$.

The May 2026 paper of Chousionis, Li, and Zhang [2] gives counterexamples for $0 < \alpha < 2$ and explicitly says that extension to $[2, 4)$ may be possible. The construction below covers all $0 < \alpha < 4$.

We use the group law

$$(x, y, t)(x', y', t') = (x + x', y + y', t + t' + \tfrac{1}{2}(xy' - yx'))$$

and the Korányi norm

$$\|(x, y, t)\| = ((x^2 + y^2)^2 + 16t^2)^{1/4}.$$

Changing the harmless normalization constant does not affect the argument.

2 Idea of the proof

The vertical coordinate of the horizontal lift of a planar graph is exactly the trapezoidal-rule error for its slope. We arrange that slope as the primitive of a bounded stopped dyadic martingale. Its increments ε_j can be square summable—which keeps the martingale bounded on most paths and the lift regular—while $\sum_j \varepsilon_j^{\alpha/2}$ diverges whenever $\alpha < 4$.

On a fixed positive proportion of pairs crossing the two children of every active dyadic interval, the current martingale increment dominates all later increments. These pair rectangles are disjoint across scales. Since K_α is nonnegative, their contributions add rather than cancel, and the K_α energy is infinite. Uniform L^2 bounds for the truncations would bound that energy, giving the contradiction.

Remark 1.6. Our good kernels are not assumed to be -1 -homogeneous, so the theorem is superficially stronger than the original result of David [19] mentioned in the first section. However, the inhomogeneous variant is well-known for odd smooth kernels in $\mathbb{R}^n$. The proof is, for example, outlined in a sequence of exercises at the end of [21, Part II]. A different proof (assuming only C^2 -regularity from k) is also contained in [62].

In the next two subsections, we will explain some further results.

1.2. Non-negative kernels. SIOs on regular curves in $\mathbb{H}$ were first studied by Chousionis and Li in [8]. The kernels $k: \mathbb{H} \setminus \{0\} \rightarrow \mathbb{C}$ considered in [8] are not "good" in the sense of Definition 1.4. Instead, they are **non-negative** -1 -homogeneous kernels of the form

$$k_\alpha(x, y, t) = \frac{(\sqrt{|t|/\|p\|})^\alpha}{\|p\|}, \quad p = (x, y, t) \in \mathbb{H} \setminus \{0\}, \alpha \geq 1.$$

Chousionis and Li proved that k_α with $\alpha \geq 8$ is a CZ kernel for regular curves $\gamma \subset \mathbb{H}$, and with Zimmerman they found a generalisation of this result to arbitrary Carnot groups [10]. Conversely, they also showed in [8] that if $E \subset \mathbb{H}$ is 1-regular, and k_2 is a CZ kernel for E , then E is contained on a regular curve. It may sound astounding that non-negative kernels could ever be CZ kernels. A heuristic explanation comes from noting that k_α vanishes identically on the plane $\{(x, y, t) : t = 0\}$. Consequently, if $\ell \subset \mathbb{H}$ is a horizontal line, then the (maximal) SIO induced by $(k_\alpha, \mathcal{H}^1|_\ell)$ is the zero operator. In contrast, our good kernels restricted to horizontal lines are odd, and the induced SIOs on horizontal lines behave like the Hilbert transform (at least when the kernels are -1 -homogeneous).

Our technique also applies to the kernels k_α by Chousionis-Li:

Theorem 1.7. *The kernels k_α are CZ kernels for regular curves in $\mathbb{H}$ for $\alpha \geq 4$.*

Recall that Chousionis and Li [8] proved this for $\alpha \geq 8$. It would be very interesting to know (as also Chousionis and Li point out) if the result persists for $\alpha \geq 2$; then we could infer that k_2 is a CZ kernel for a 1-regular set $E \subset \mathbb{H}$ if and only if E is contained on a regular curve. We close the section with another open question. While our technique applies to the kernels k_α , our main result, Theorem 1.5 does not. So, we ask for a class of kernels which simultaneously contains odd and horizontally odd kernels, and the non-negative kernels of Chousionis-Li. Here is one suggestion (*caveat emptor!*):

Question 1. *Let $k: \mathbb{H} \setminus \{0\} \rightarrow \mathbb{C}$ be a smooth -1 -homogeneous function which is a CZ kernel for horizontal lines, with uniform constants. Is k then a CZ kernel for regular curves?*

After the first version of this paper was posted on the *arXiv*, Chousionis, Li, and Zimmerman [9] established the following partial result in all Carnot groups: whenever a 1-dimensional standard kernel (see Definition 2.1) is a CZ kernel for all horizontal lines, with uniform constants, then it is also a CZ kernel for regular $C^{1,\alpha}$ -curves for $\alpha > 0$.

1.3. SIOs on Lipschitz flags. The L^2 -boundedness of 1-codimensional SIOs on Lipschitz surfaces is a key component in the method of layer potentials. This is a powerful

Figure 1: Theorem 1.7 and the question in Fässler–Orponen [1], PDF page 4.

3 A stopped-martingale energy lemma

For $g \in L^\infty[0, 1]$ and $y < x$, write

$$\mathcal{E}_g(x, y) := \int_y^x \left(\frac{x+y}{2} - s \right) g(s) ds.$$

Lemma 1. *Let $0 < q < 2$. There are a bounded function g on $[0, 1]$, decreasing numbers $\varepsilon_j > 0$, dyadic levels $n_j = 8j$, and sets A_j which are unions of dyadic intervals of level $n_j - 1$, such that*

$$|A_j| \geq \frac{3}{4}, \quad \sum_j \varepsilon_j^2 < \infty, \quad \sum_j \varepsilon_j^q = \infty.$$

Moreover, if $I = [a, a + \ell] \subset A_j$ is one of those level- $(n_j - 1)$ intervals, then

$$|\mathcal{E}_g(x, y)| \geq c_* \varepsilon_j \ell^2 \tag{3.1}$$

whenever

$$y \in [a, a + \ell/8], \quad x \in [a + 7\ell/8, a + \ell],$$

where $c_* > 0$ is absolute. All the resulting pair rectangles, over every I and j , are mutually disjoint.

Proof. Choose

$$\frac{1}{2} < p < \frac{1}{q}, \quad \varepsilon_j = cj^{-p},$$

where $c > 0$ is small enough that $4 \sum_j \varepsilon_j^2 \leq 1/4$. The two asserted series then have the required behavior.

Let $\mathcal{F}_n$ be the dyadic filtration on $[0, 1)$, and let r_n be the n th Rademacher function: it is $+1$ and -1 on the two children of every level- $(n-1)$ interval. Starting with $M_0 = 0$ and $A_1 = [0, 1)$, define

$$d_j = \varepsilon_j \mathbf{1}_{A_j} r_{n_j}, \quad M_j = M_{j-1} + d_j, \quad A_{j+1} = A_j \cap \{|M_j| < 1/2\}.$$

This is a stopped martingale. Doob's inequality gives

$$|A_j^c| \leq \mathbb{P} \left(\max_{k < j} |M_k| \geq \frac{1}{2} \right) \leq 4\mathbb{E}|M_{j-1}|^2 \leq 4 \sum_k \varepsilon_k^2 \leq \frac{1}{4}.$$

The martingale converges almost everywhere and in L^2 to a function g . It is bounded: after the first crossing of $1/2$ the process stops, and the overshoot is at most ε_1 ; a path which never crosses stays below $1/2$.

Fix an active parent I at stage j . Earlier details are constant on I , and $\mathcal{E}$ annihilates constants. After rescaling I to $[0, 1]$, the contribution of the current Haar detail to $\mathcal{E}_g(x, y)$ has absolute value

$$\varepsilon_j \ell^2 \frac{(2X - 1)(1 - 2Y)}{4} \geq \frac{9}{64} \varepsilon_j \ell^2, \quad (3.2)$$

where $Y \in [0, 1/8]$ and $X \in [7/8, 1]$.

It remains to control descendants. A later detail d_k has mean zero on each of its dyadic parent cells J , whose length is $2^{-8(k-j)}\ell$. On a cell wholly contained in $[y, x]$, subtract a constant from the affine weight $(x + y)/2 - s$; its oscillation on J is at most $|J|$. The full cells therefore contribute in total at most $\varepsilon_k \ell |J|$. The two possible boundary cells obey the same bound up to a factor two. Consequently

$$\left| \int_y^x \left(\frac{x + y}{2} - s \right) d_k(s) ds \right| \leq 3\varepsilon_k \ell^2 2^{-8(k-j)}.$$

Since $\varepsilon_k \leq \varepsilon_j$, the complete descendant tail is at most

$$\frac{3}{2^8 - 1} \varepsilon_j \ell^2 < \frac{1}{2} \cdot \frac{9}{64} \varepsilon_j \ell^2.$$

This proves (3.1), for example with $c_* = 9/64 - 3/255$.

Finally, a selected pair lies in opposite immediate children of I . Thus I is its unique smallest dyadic common ancestor, which proves disjointness over all parents and stages. $\square$

4 The regular curve and divergent energy

Theorem 2. *For every $0 < \alpha < 4$, there is a compact regular curve $\Gamma_\alpha \subset \mathbb{H}$ such that the truncations of the singular integral associated with K_α are not uniformly bounded on $L^2(\mathcal{H}^1|_{\Gamma_\alpha})$.*

Proof. Set $q = \alpha/2 < 2$ and take g from Lemma 1. Extend g by zero outside $[0, 1]$, and define

$$f(x) = \int_0^x g(u) du, \quad F(x) = \int_0^x f(u) du.$$

Since g is bounded, f is Lipschitz. Consider the horizontal graph map

$$\Phi(x) = \left(x, f(x), -F(x) + \frac{xf(x)}{2} \right). \quad (4.1)$$

For $h = x - y$, direct use of the group law gives

$$\Phi(y)^{-1}\Phi(x) = (h, f(x) - f(y), -\mathcal{E}_g(x, y)), \quad (4.2)$$

because integration by parts yields

$$F(x) - F(y) - \frac{h}{2}(f(x) + f(y)) = \int_y^x \left(\frac{x+y}{2} - s \right) g(s) ds.$$

The same identity and boundedness of g show

$$|f(x) - f(y)| \lesssim |h|, \quad |\mathcal{E}_g(x, y)| \lesssim |h|^2.$$

Hence

$$|x - y| \leq d(\Phi(x), \Phi(y)) \lesssim |x - y|.$$

Thus Φ is bi-Lipschitz, and $\Gamma_\alpha := \Phi([0, 1])$ is a compact regular curve. Its Hausdorff measure is comparable, under Φ , to Lebesgue measure.

Equation (4.2) and the upper metric estimate imply

$$K_\alpha(\Phi(y)^{-1}\Phi(x)) \gtrsim \frac{|\mathcal{E}_g(x, y)|^{\alpha/2}}{|x - y|^{\alpha+1}}. \quad (4.3)$$

For a pair in the rectangle associated with an active interval I of length ℓ at stage j , Lemma 1 and $|x - y| \leq \ell$ give

$$K_\alpha(\Phi(y)^{-1}\Phi(x)) \gtrsim \frac{\varepsilon_j^{\alpha/2}}{\ell}.$$

The pair rectangle has planar measure $\ell^2/64$, so its contribution to the positive double integral is at least $c\varepsilon_j^{\alpha/2}\ell$. The rectangles are disjoint, and the total length of the active parents is $|A_j| \geq 3/4$. Consequently

$$\iint_{[0,1]^2} K_\alpha(\Phi(y)^{-1}\Phi(x)) dx dy \gtrsim \sum_j \varepsilon_j^{\alpha/2} |A_j| = \infty. \quad (4.4)$$

To phrase the last step directly in terms of truncations, retain only the first N generations of pair rectangles and choose the truncation radius smaller than their least metric separation. If the truncations were uniformly L^2 bounded, positivity and Cauchy–Schwarz, applied to the indicator of Γ_α , would bound every such partial double integral by a constant times $\mathcal{H}^1(\Gamma_\alpha)$. The lower bounds in (4.4) tend to infinity. This contradiction proves the theorem. $\square$

5 Consequences, limitations, and verification

Theorem 2 answers the source’s $\alpha \geq 2$ question negatively, including every previously unsettled value $2 \leq \alpha < 4$. Combined with Fässler–Orponen’s positive theorem for $\alpha \geq 4$, it shows that four is the exact threshold.

This does *not* by itself settle the separate smooth-kernel Question 1 shown at the bottom of Figure 1. The nonsmooth behavior of K_α across the horizontal plane is essential to membership in this particular family.

The finite checker supplied with the packet verifies the exact Haar formula, the exponent window, the variance budget, and finite trapezoidal lower bounds. It is not used in the proof. A bounded novelty search used the exact formula, question text, source title and authors, and the terms “nonnegative kernel”, “regular curve”, and “ $\alpha = 4$ ”. It found [2], which handles only $0 < \alpha < 2$ and explicitly flags $[2, 4)$ as a possible extension; no primary source stating the result above was found.

Human review should concentrate on the descendant estimate in Lemma 1 and on the Hausdorff-measure transfer in (4.4).

References

- [1] K. Fässler and T. Orponen, *Singular integrals on regular curves in the Heisenberg group*, arXiv:1911.03223; J. Math. Pures Appl. 153 (2021), 30–154.
- [2] V. Chousionis, S. Li, and L. Zhang, *On singular integrals with non-negative kernels in the Heisenberg group*, arXiv:2605.17680 (2026).